

AP Precalc Unit 1 Wkst A
Linear & Angular Speed

Name: _____

For ALL of the following problems you must a) draw a picture, and b) solve the problem. Be sure to write your answer with the units represented.

- 1) Find the radian measure of the central angle of a circle with a radius of 6 feet and an intercepted arc of 27 feet.

- 2) Find the radian measure of the central angle of a circle with a radius of 25 centimeters and an intercepted arc of 10 centimeters.

For the following questions find the length of an arc that is intercepted by the given central angle in a circle with the given measure.

3) central angle of 120° and radius of 9 miles.	5) central angle of $\frac{11\pi}{6}$ and radius of 18 millimeters.
4) central angle of 50° and radius of 3 centimeters.	6) central angle of $\frac{3\pi}{4}$ and <u>diameter</u> of 10 feet.

7.) Find the angular and linear velocity (speed) of a gear with radius of 10 centimeters if one revolution takes 14 seconds.

8.) Find the angular and linear velocity (speed) of a wheel with radius of 35 inches if half a revolution takes 70 seconds.

9.) Find the angular and linear velocity (speed) of a gear with a diameter of 4 meters if it makes 5 revolutions per minute.

10) Find the angular and linear velocity (speed) of a wheel with radius 120 feet if it rotates 240° in 30 seconds.

11.) Kelly, Andrew, Cheri, and Hayden are headed to Florida for Spring Break. They are traveling in a car that has 24 inch tires. How fast are they traveling if 2 revolutions take 1.5 seconds? Express your answer in miles per hour. Give both the exact answer and the answer rounded to 1 d.p.